

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA0718

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Software: Cisco Packet Tracer, Wireshark

QUESTIONS: 5

Total Marks:50

Annexure A

Debugging & Optimisation Task

Code of Conduct:

*I P. Sai Charan Tej(192465048) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, wellstructured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

Question 1 – IPv4 Subnetting Debugging

Given configuration:

- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (/26)
- Default Gateway: 192.168.10.1

For /26, the block size is $256 - 192 = 64$. Therefore, the configured IP 192.168.10.65 belongs to the 192.168.10.64/26 subnet.

Network Address : 192.168.10.64
Broadcast Address : 192.168.10.127
Valid Host Range : 192.168.10.65 – 192.168.10.126
Usable Hosts : $2^6 - 2 = 62$

The gateway 192.168.10.1 belongs to 192.168.10.0/26, so it is outside the workstation's subnet.

Example corrected configuration:

IP Address : 192.168.10.66
Subnet Mask : 255.255.255.192
Default Gateway : 192.168.10.65

Error Identified: The default gateway is outside the workstation's /26 subnet.

Error Solved: The workstation is given a gateway from the same 192.168.10.64/26 subnet. The corrected subnet provides 62 usable host addresses.

Question 2 – IPv6 Address Debugging

Given: Device A = 2001:db8::ff00:42:8329/64; Device B = 2001:db8:0:0:1::1/64.

Device A: 2001:0db8:0000:0000:0000:ff00:0042:8329
Device B: 2001:0db8:0000:0001:0000:0000:0000:0001

Device A prefix: 2001:db8:0:0::/64
Device B prefix: 2001:db8:0:1::/64

The devices are in different /64 subnets.

Example corrected configuration:

Device A: 2001:db8::ff00:42:8329/64
Device B: 2001:db8::1/64
Corrected subnet: 2001:db8::/64
/64 address space: $2^{64} = 18,446,744,073,709,551,616$ addresses

Error Identified: Device A and Device B use different /64 network prefixes.

Error Solved: Device B is assigned an address within the same 2001:db8::/64 subnet as Device A.

Question 3 – VLSM Subnet Allocation Optimization

The network 192.168.1.0/24 is divided using /26, /27, and /28. The assessment states that some departments have insufficient addresses while others have unused addresses.

Prefix	Total Addresses	Usable Hosts
/26	64	62
/27	32	30
/28	16	14

/26 Network: 192.168.1.0 Broadcast: 192.168.1.63
Hosts: 192.168.1.1 – 192.168.1.62

/27 Network: 192.168.1.64 Broadcast: 192.168.1.95
Hosts: 192.168.1.65 – 192.168.1.94

/28 Network: 192.168.1.96 Broadcast: 192.168.1.111
Hosts: 192.168.1.97 – 192.168.1.110

Recommended VLSM approach:

- Determine the host requirement of each department.
- Arrange departments from largest to smallest requirement.
- Allocate the largest suitable subnet first.
- Assign smaller subnets to departments with smaller requirements.
- Use remaining address space efficiently.

Error Identified: Poorly matched fixed-size subnet allocation can waste addresses in some departments and leave others short of addresses.

Error Solved: Use VLSM according to actual departmental requirements. Exact unused-address counts cannot be calculated because the question does not provide individual department host requirements.

Question 4 – IPv6 Configuration Debugging

Given

- **Device A:** 2001:db8::ff00:42:8329/64
- **Device B:** 2001:db8:0:0:1::1/64

Both devices are connected to the same network, but communication fails. The assessment asks you to expand the addresses, determine the prefixes, verify the subnet, identify the mismatch, optimize the configuration, and calculate the available addresses.

Step 1: Expand the IPv6 Addresses

Device A:

2001:db8::ff00:42:8329/64

Complete notation:

2001:0db8:0000:0000:0000:ff00:0042:8329/64

Device B:

2001:db8:0:0:1::1/64

Complete notation:

2001:0db8:0000:0000:0001:0000:0000:0001/64

Step 2: Determine Network Prefix

For /64, the first 64 bits represent the network prefix.

Device A:

2001:db8:0000:0000::/64

Device B:

2001:db8:0000:0000::/64

Step 3: Verify the Subnet

Both devices have the same /64 prefix:

2001:db8:0000:0000::/64

Therefore:

Device A → Same subnet

Device B → Same subnet

So, there is actually no IPv6 prefix mismatch in the addresses given in Question 4.

Step 4: Identify the Error

The addresses themselves belong to the **same /64 subnet**. Therefore, the supplied addressing does **not** explain the communication failure.

Possible causes that need to be checked are:

- Incorrect IPv6 interface configuration
- Interface/link being down
- Incorrect IPv6 routing
- Firewall/ACL blocking IPv6 traffic
- Duplicate IPv6 address

- Incorrect gateway or neighbor configuration

Step 5: Optimized Configuration

A simple valid configuration can be:

Device A:

IPv6 Address : 2001:db8::ff00:42:8329/64

Device B:

IPv6 Address : 2001:db8::1/64

Network:

2001:db8::/64

Both addresses are within the same /64 network.

Step 6: Available Host Addresses

For an IPv6 /64 network:

$$2^{(128 - 64)}$$

$$= 2^{64}$$

$$= 18,446,744,073,709,551,616$$

So the subnet provides **2^{64} possible interface addresses**.

Error Identified

The given IPv6 addresses do not have a prefix mismatch. Both devices belong to the same 2001:db8::/64 subnet. Therefore, the communication failure cannot be attributed to the IPv6 subnet addressing shown in the question.

Error Solved

Verified that both devices are correctly within the same /64 subnet and provided an optimized address example. Further debugging should focus on interfaces, IPv6 routing, neighbor discovery, or filtering configuration.

Question 5 – Routing Table Debugging

Given routing table:

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Destination IP: 192.168.2.100. It belongs to 192.168.2.0/24, so the longest-prefix matching route selects next hop 10.0.0.3.

Destination	: 192.168.2.100
Matching route	: 192.168.2.0/24
Selected next hop	: 10.0.0.3
Default route	: 10.0.0.1

Error Identified: Based only on the supplied routing table, there is no incorrect route for 192.168.2.100. The correct longest-prefix route already points to 10.0.0.3.

Error Solved: The correct route and next-hop selection were verified. If packets still fail, next-hop reachability, interface status, and the return route should be checked.

7. Final Error Summary

Question	Error Identified	Error Solved / Correction
Q1	Gateway 192.168.10.1 is outside the PC's /26 subnet.	Use a gateway in 192.168.10.64/26; 62 usable hosts.
Q2	IPv6 devices use different /64 prefixes.	Assign both devices to 2001:db8::/64.
Q3	Inefficient subnet allocation.	Use VLSM based on departmental requirements.
Q4	IPv6 prefix mismatch.	Correct Device B so both devices use the same /64 subnet.
Q5	No routing-table error is evident for 192.168.2.100.	Longest-prefix match correctly selects 10.0.0.3.

Conclusion

The debugging process identifies the key addressing and routing issues and provides appropriate corrections. IPv4 subnet validation, IPv6 prefix matching, VLSM, and longest-prefix routing analysis help ensure a reliable and efficient network configuration.

